

Table 1. Overview of domain truncation, CBM substitution and linker studies in LPMOs.

An updated version of this table is available online at <https://github.com/gcourtade/papers/tree/master/2022/LPMO-modularity-review>

Enzyme (regioselectivity, substrate specificity)	Modular organization of wild-type (WT) and truncation variants	(a) Key findings	References
Vc LPMO10B (GbpA) (C1-oxidizing, chitin)	AA10-GbpA2-GbpA3-CBM73 (WT); AA10; AA10-GbpA2-GbpA3; GbpA2-GbpA3; CBM73	Chitin-binding is mainly aided by the CBM and to a lesser extent by the LPMO domain. The LPMO domain is required for mucin binding and GbpA2 and GbpA3 in combination with the LPMO domain are important for intestinal colonization in a cholera mouse model. • In-depth characterization of full-length and truncated variants of Vc GbpA and a homologue from <i>V. campbellii</i> (Vh GbpA) showed that the catalytic LPMO domains exhibit properties similar to natural single-domain LPMOs. GbpA2 and GbpA3 affect chitin binding and degradation. GbpA3 has evolved to interact with the reduced catalytic copper site to prevent off-pathway reactions in the absence of substrate; substrate binding by CBM73 weakens this interaction, enabling activation only when substrate is present.	Wong et al. 2012 PLoS Pathog ; Zhou et al. 2026 Biochemistry
Sc LPMO10C (C1-oxidizing, cellulose)	AA10-CBM2 (WT); AA10	Loss of cellulose binding affinity for the CBM truncated enzyme. • Reduced cellulose activity of CBM truncated enzyme. • NMR spectroscopy study showing structural and dynamic features of a modular LPMO. • The substrate-binding affinity resides with the CBM. • The CBM is beneficial for LPMO activity and operational stability at lower substrate concentrations and promotes localized and repeated oxidation of the substrate. • Truncation of the CBM leads to elevated H ₂ O ₂ production and decreased enzyme stability (both in absence and presence of cellulose). Release of copper by damaged enzymes promotes H ₂ O ₂ production, which increases LPMO catalytic rate and inactivation. • Observed synergistic effects when combining the two enzyme forms are due to a combination of high oxidase activity (i.e., increased LPMO-dependent H ₂ O ₂ production) by the truncated enzyme and efficient productive use of H ₂ O ₂ (i.e., peroxxygenase activity) by the full-length enzyme. • The linker adopts an extended conformation functioning as a flexible tether allowing independent domain motions. Linker variants with altered proline/glycine/serine content showed that linker conformation affects both catalytic performance and enzyme stability; a more compact linker conformation reduces operational stability. • Fluorescent labeling of C1-oxidized sites combined with confocal microscopy showed that the CBM determines the distribution of oxidized positions on the cellulose surface: the full-length enzyme produces a punctate (localized) oxidation pattern, whereas the CBM-free variant produces a dispersed pattern. The CBM impact is dependent on both cellulose concentration and cellulose type. • The CBM2 possesses a copper-binding site (Met-Met-His motif) on the opposite side from the cellulose-binding surface that confers redox protection by	Forsberg et al. 2014 JBC ; Forsberg et al. 2014 PNAS ; Courtade et al. 2018 JBC ; Stepnov et al. 2022 Sci Rep ; Forsberg et al. 2023 JBC ; Raji et al. 2023 Cellulose ; Forsberg et al. 2025 JACS

		preferentially binding free Cu(I) and interacting with the reduced catalytic copper site. This interdomain copper site prevents off-pathway reactions in the absence of substrate; the protective interaction is relieved upon cellulose binding.	
Nc LPMO9C (C4-oxidizing cellulose, cello-oligosaccharides, xyloglucan)	AA9-CBM1 (WT); AA9	K _d measured for phosphoric acid swollen cellulose (PASC) and xyloglucan showed weaker binding for CBM truncated Nc LPMO9C. • No CBM truncation effect on activity on PASC but a 2-fold reduction in the rate of xyloglucan degradation. • Truncation of the CBM reduced the binding affinity and LPMO activity but did not affect regioselectivity. • The linker is important for thermal stability. • The single-domain variant oxidized cellulose fibers to a higher extent than its CBM-containing natural variant and released less soluble products, indicating a more dispersed oxidation pattern without a CBM. Activity on the Cellulose II allomorph was almost exclusively detected for LPMOs containing a CBM1. • The CBM improved binding to amorphous (PASC), natural (Cell I), and alkali-treated (Cell II) cellulose, and was essential for significant binding of the non-reduced LPMO to Cell I and Cell II. Sequential fiber dissolution analyses showed that cuts by the CBM-free enzyme are more evenly spread through the fiber and that the truncated enzyme can penetrate deeper into the fiber core. • CBM1 promoted cellulose-binding affinity but reduced oxidase activity. The effect of CBM1 on monooxygenase activity was linked to the reductant type and cellulose crystallinity. Linker truncation enhanced cellulose-binding affinity but decreased oxidase and peroxidase activities.	Borisova et al. 2015 JBC ; Laurent et al. 2019 IJMS ; Støpamo et al. 2024 Carbohydr Polym ; Støpamo et al. 2024 Biotechnol Biofuels Bioprod ; Zhao et al. 2024 IJMS
Cf LPMO10 (C1/C4-oxidizing, cellulose and C1-oxidizing, chitin)	AA10-CBM2 (WT); AA10; <i>Tb</i> AA10-CBM2; AA10-CBM3; AA10-CBM10	Study on deleting and replacing CBMs in two cellulose-oxidizing LPMOs. Introduction of other types of cellulose binding CBMs both potentiated and inhibited the LPMO activity. Such effects were both enzyme and substrate specific. • Changed ratios between non-oxidized and oxidized products when replacing the CBM2 by a CBM10 → CBMs can modulate the mode of action of LPMOs.	Crouch et al. 2016 JBC
Tb LPMO10 (C1-oxidizing, cellulose)	AA10-CBM2 (WT); AA10; <i>Cf</i> AA10-CBM2; AA10-CBM3; AA10-CBM10	See above (<i>Cf</i> LPMO10).	Crouch et al. 2016 JBC
Cj LPMO10A (C1-oxidizing, chitin)	AA10-CBM5-CBM73 (WT); AA10; AA10-CBM5	Removal of both CBMs reduced LPMO activity toward α -chitin compared with the full-length enzyme, but in synergistic reactions with an endo-chitinase equal levels of solubilized products were observed. • Structural analysis of two similar chitin-binding CBMs with different affinity for crystalline chitin and soluble chitohexaose. The effect of CBMs on chitin oxidation is substrate concentration dependent; at low concentrations, the CBM-containing variants performed better, whereas at high concentrations the differences were less apparent.	Forsberg et al. 2016 JBC ; Madland et al. 2021 JBC
Hj LPMO9A (C1/C4-oxidizing cellulose)	AA9-CBM1 (WT); 21 amino acid linker AA9-fragment; AA9 (including three mutated variants; Y24A, Y211A &	Removal of the CBM, post-translationally by papain hydrolysis, led to a truncated variant with 21 remaining residues of the predicted linker which exhibited reduced binding and activity towards cellulose compared to the full-length enzyme. The X-ray structure	Hansson et al. 2017 JBC ; Danneels et al. 2019 Biotechnol Bioeng

	Y24A_Y211A)	revealed that the glycosylated linker forms an integral part covering a hydrophobic patch on the catalytic LPMO domain. • Removing the CBM resulted in reduced binding but did not alter the oxidative regioselectivity. However, the effects of point mutations in the catalytic domain on oxidative regioselectivity became more apparent in the absence of the CBM.	
Tf LPMO10B (C1-oxidizing, cellulose)	AA10-FnIII-CBM2 (WT); AA10; AA10-FnIII; AA10-CBM2	Binding is mediated mainly by the CBM and to some extent by the LPMO domain. Removal of the FnIII-like domain (called X1) had no effect on binding nor on activity.	Kruer-Zerhusen et al. 2017 Biotechnol Biofuels
Ma LPMO10B (C1/C4-oxidizing, cellulose and C1-oxidizing, chitin)	AA10-CBM2 (WT); AA10	Deletion of the CBM affected the operational stability of the LPMO but did not affect the ratio of regioselective C1:C4 oxidation.	Forsberg et al. 2018 JBC
Bc LPMO10A (C1-oxidizing, chitin)	AA10-FnIII-FnIII-CBM5 (WT); AA10; AA10-FnIII; AA10-FnIII-FnIII	Enzyme functionality was strongly dependent on the CBM that is responsible for substrate binding and protects the enzyme from autocatalytic inactivation. Truncation of one or two of the FnIIIs (both in combination with the CBM) resulted in essentially the same effect as when only the CBM was removed.	Mutahir et al. 2018 FEBP
Bt LPMO10A (C1-oxidizing, chitin)	AA10-FnIII-FnIII-CBM5 (WT); AA10; CBM5; AA10-FnIII-CBM5; AA10-CBM5; FnIII-FnIII	The CBM is essential for binding to α - and β -chitin. The FnIII-like domains do not have a role in chitin-binding.	Manjeet et al. 2019 Int J Biol Macromol
Pa LPMO9H (C1/C4-oxidizing cellulose, cello-oligosaccharides, xyloglucan)	AA9-CBM1 (WT); AA9	Truncation of the CBM weakened substrate binding and affected the catalytic performance on nanofibrils, amorphous and crystalline cellulosic substrates, but there was no effect on the activity on cellobiose. • Increasing the substrate concentration reduces the need for a CBM. • The truncated variant showed a modified regioselectivity with increased C1-oxidation. • Optical and atomic force microscopy of the insoluble fraction revealed that both variants can promote disruption of the cellulose network and that the CBM is not essential.	Chalak et al. 2019 Biotechnol Biofuels
Jd LPMO10A (C1-oxidizing, chitin)	AA10-CBM5-GH18 (WT); AA10; AA10-CBM5; CBM5-GH18; GH18	Synergy study that showed intramolecular synergy between the LPMO domain and the chitinase (GH18) domain. Comparison of the chitinolytic efficiency of the full-length enzyme and combinations of truncated variants showed that the full-length enzyme is more efficient compared to any combination of its separately produced domains.	Mekasha et al. 2020 JBC
Mt LPMO9B (C1-oxidizing, cellulose)	AA9-CBM1 (WT); AA9	The CBM promotes cellulose degradation in the full-length enzyme but does not affect oxidative regioselectivity.	Sun et al. 2021 ACS Sustain Chem Eng
Bc LPMO9C (Unknown regioselectivity, cellulose)	55 AA AA9-CBM1 linker (WT); 44 AA AA9-CBM1 linker; 18 AA AA9-CBM1 linker; 7 AA linker AA9-CBM1; AA9	Studies on linker truncation showed that shortening the linker or removing the CBM reduces substrate binding.	Srivastava et al. 2022 Microbiol Spectr
Cfla LPMO10D (C1-oxidizing, β -chitin)	AA10-CBM2 (WT); AA10	• A chitin-active LPMO from <i>Cellulomonas flavigena</i> , strictly active on β -chitin. • The CBM2 mediates enzyme localization to substrates and prolongs LPMO activity.	Li et al. 2022 Appl Environ Microbiol
Mt LPMO (AA13) (starch-active)	AA13 (WT, single-domain); AA13-Cn_CBM;	• Four CBMs from different organisms were fused to a single-domain starch-active AA13 LPMO. • CBM chimeras showed	Zhang et al. 2023 J Agric Food Chem

	AA13-_Pv_CBM; AA13-_Ac_CBM; AA13-_Cv_CBM	enhanced substrate binding, H ₂ O ₂ production (up to 4.6-fold), and oxidative product yields (up to 6.0-fold for amylopectin). • CBM effects were starch-type dependent.	
Ao LPMO9A (C1/C4-oxidizing, cellulose)	AA9 (WT, single-domain); AA9-CBM1 (fused)	• CBM1 fusion reduced oxidase activity while promoting cellulose binding. • CBM1 improved monooxygenase activity on high-, medium-, and low-crystallinity celluloses with ascorbic acid as electron donor.	Zhao et al. 2024 IJMS
Ao LPMO9B (C4-oxidizing, cellulose)	AA9 (WT, single-domain); AA9-CBM1 (fused)	See above (Ao LPMO9A). • C4-oxidizing LPMOs showed lower oxidase and peroxidase activities compared to C1/C4-oxidizing LPMOs.	Zhao et al. 2024 IJMS
Ao LPMO9C (C1/C4-oxidizing, cellulose)	AA9-CBM1 (WT); AA9 (truncated)	• CBM1 truncation reduced cellulose-binding affinity. • CBM1 effects on monooxygenase activity depended on reductant type and cellulose crystallinity.	Zhao et al. 2024 IJMS
Mt LPMO9G (C1/C4-oxidizing, cellulose)	AA9-CBM1 (WT); AA9 (truncated)	• CBM truncation reduced substrate binding affinity and enzymatic activity. • CBM fusion to single-domain <i>Mt</i> LPMO9L (see below) improved binding, activity, and H ₂ O ₂ tolerance. • CBM promotes proximity of active site to substrate, leading to more efficient H ₂ O ₂ consumption and reduced auto-oxidative damage.	Gao et al. 2024 JBC
Mt LPMO9L (C1-oxidizing, cellulose)	AA9 (WT, single-domain); AA9-CBM1 (fused)	• Fusion of a CBM to the naturally single-domain LPMO substantially improved substrate binding affinity, enzymatic activity, and H ₂ O ₂ tolerance. • Molecular dynamics simulations confirmed that CBM fusion enhances proximity of the active site to the substrate, promoting multilocal cleavage and reducing exposure of the copper active site to H ₂ O ₂ .	Gao et al. 2024 JBC
Gt LPMO9A-2 (cellulose)	AA9-CBM104 (WT); CBM104 (isolated); AA9	• CBM104 is a new CBM family at the C-terminus of a brown-rot fungal LPMO9. • CBM104 binds selectively to native crystalline cellulose (cellulose I) but not to amorphous or artificially modified crystalline cellulose, unlike CBM1 which binds all cellulosic materials. • Adsorption efficiency of CBM104 to cellulose I was >20-times higher than CBM1. • CBM104 binds to hydrophilic regions of cellulose microfibrils, while CBM1 recognizes the hydrophobic surface. • CBM104s appended to different catalytic domains all bind cellulose I, and two conserved disulfide bonds fixing a pair of α -helices are essential for binding.	Kojima et al. 2025 Carbohydr Polym ; Kojima et al. 2025 J Appl Glycosci
Cc LPMO10-Man5 (C1-oxidizing, cellulose)	AA10-GH5-CBM3-CBM3 (WT); AA10-GH5; AA10; GH5-CBM3-CBM3; other truncations	• A multi-catalytic modular enzyme from <i>Caldibacillus cellulosovorans</i> combining LPMO and mannanase activities. • The LPMO domain is incapable of accessing softwood cellulose in the absence of the mannanase domain. • Intramolecular coupling of LPMO with mannanase and CBMs ensures that the LPMO is directed to areas where mannans are removed and cellulose becomes available. • Synergy was most pronounced with cellobiohydrolases.	Forsberg et al. 2025 FEBS J
Ec LPMO (C1-oxidizing, chitin)	AA10-DUF_A-DUF_B-CBM73 (WT); AA10-DUF_A-DUF_B (Δ CBM73); AA10 (catalytic domain)	• A tetra-modular LPMO from <i>Enterobacter cloacae</i> . • The CBM73 enhances chitin binding and promotes synergy with endogenous chitinases (up to 14-fold and 60-fold increase in GlcNAc release from α - and β -chitin, respectively). • Deletion of both DUFs reduced LPMO activity. • DUF-B contains two conserved histidines that coordinate the AA10-bound copper, forming a previously	Bardhan et al. 2026 JBC

		unreported "inter-domain tetra-histidine copper coordination" center that may protect the LPMO active site from oxidative damage.	
Pa AA9E-CBM1 (C1-oxidizing, cellulose)	<i>Pa</i> AA9E-CBM1 (WT); <i>Pa</i> AA9E-HFB2; <i>Pa</i> AA9E-HFB4; <i>Pa</i> AA9E-PHA-Bd	• The native CBM1 was swapped with hydrophobins (HFB2, HFB4) or the PHA-binding domain of a bacterial PHB depolymerase (PHA-Bd). • Binding was pH-dependent; swapping of CBM1 did not significantly alter thermal stability, and chimeras retained wild-type-like cellulose activity. • <i>Pa</i> AA9E-PHA-Bd showed up to 66-fold higher binding to PHA than the parental enzyme, shifting substrate specificity from cellulose to PHA (and PLA, PBS, PET, PE). • <i>Pa</i> AA9E-PHA-Bd consumed H₂O₂ in a PHA-dependent manner and modified PHA films (holes of 2–15 μm), providing evidence of LPMO-dependent peroxygenase activity on a synthetic polymer.	Munzone et al. 2024 Chem Bio Eng
Ls AA9A (C4-oxidizing, cellulose/cello-oligosaccharides)	<i>Ls</i> AA9A (WT, single-domain); <i>Ls</i> AA9A-CBM1 (fused)	• CBM1 from <i>Pa</i> AA9E was fused to the naturally single-domain <i>Ls</i> AA9A. • CBM1 fusion improved binding to cellulose at pH $\geq$ 6 (up to 1.3-fold at pH 8.0) and improved binding to the polyesters PHA, PBS, and PE.	Munzone et al. 2024 Chem Bio Eng
Mo PMO9A (C4-oxidizing, mixed (1 $\rightarrow$ 3,1 $\rightarrow$ 4)- β -D-glucans)	AA9-DUF-CBM (WT); AA9-DUF; AA9	• A unique AA9 LPMO from the plant pathogen <i>Magnaporthe oryzae</i> that is NOT active on cellulose but active on cereal-derived mixed-linkage β-glucans. • The DUF (domain of unknown function) is essential for activity — truncation removing the DUF abolished activity. • The CBM binds chitin, not cellulose. • C4-oxidizing. • Gene deletion results in reduced pathogenicity, supporting a role in β-glucan degradation during plant infection.	Martinez-D'Alto et al. 2023 PNAS

(a) Note that the more quantitative statements in this Table need to be read with caution, because the quality of the underlying kinetic analyses varies. For example, especially in early studies, the impact of autocatalytic LPMO inactivation, leading to non-linear progress curves, was not always considered ([Eijsink et al. 2019 Biotechnol Biofuels](#)).